

Yi-Han Iris Yin

	Pokfulam Road, Hong Kong, China
	+86 151-0520-2182
	iris.yh.yin@connect.hku.hk
	https://irishellenyin.github.io/irisyin.github.io/
	orcid.org/0000-0002-5596-5059

I am currently a first-year PhD student at The University of Hong Kong majoring in astrophysics. My primary research interest centers on gamma-ray burst (GRB) central engines and their associated astrophysical phenomena. The focus of my recent researches has been on analyzing and interpreting GRB observations, especially of the peculiar ones, such as GRB 211211A, GRB 230307A and GRB 231115A. In addition, I am actively involved in studying fast X-ray transients detected by the Einstein Probe Mission.

EDUCATION

2025 – PRESENT	PhD student in Astrophysics Department of Physics <i>The University of Hong Kong</i>
2023 – 2025	M.S. in Astronomy School of Astronomy and Space Science <i>Nanjing University</i>
2019 – 2023	B.S. in Physics School of Physics <i>Nanjing University</i>

RESEARCH INTEREST

- Fast X-ray transients
- Physical origins and progenitors of GRBs
- Central engines and radiation mechanisms of GRBs
- Data and observation-oriented research on GW-related astrophysical phenomena

RESEARCH EXPERIENCE

Einstein Probe Mission

MARCH 2023 – PRESENT

Transient Advocate / Former Associated STP Member

The Einstein Probe is a mission of the Chinese Academy of Sciences dedicated to time-domain high-energy astrophysics. Its first detection of a bright X-ray transient, EP240219a, is confirmed to be an untriggered GRB (Yin et al., 2024, ApJL). I serve as the EP Transient Advocate and involved in the data analysis and discussion of many LEIA/EP-detected fast X-ray transients, e.g., GRB 230307A/LXT230307a (Sun et al., 2024, National Science Review, participant), EP240315a (Liu et al., 2025, Nature Astronomy, participant), and EP250704a (Li et al., under review).

The Hong Kong Institute for Astronomy and Astrophysics

NOV 2025 – PRESENT

Member

Created by Prof. Bing Zhang, The Hong Kong Institute for Astronomy and Astrophysics (HKIAA) is an international center of research excellence in astronomy and astrophysics and a platform for fostering communication and collaboration within the international astronomical community.

NJU GRB Group

JAN 2022 – SEP 2025

Member

As a member of Professor Bin-Bin Zhang’s GRB research team, I have participated in several research projects and am keen to discover new and exciting GRB events. Specifically, I have been deeply involved with the data analysis and follow-up GW-related research of GRB 211211A (Yin et al., 2023, ApJL), which is a peculiar long-duration burst with kilonova emission (Yang et al., 2022, Nature, participant). Moreover, I led a project to analyze MGF GRB 231115A and confirmed its origin as the second extragalactic magnetar giant flare with a Comptonized fireball model (Yin et al., 2024, ApJL).

DEC 2020 – SEP 2025

GRID Project
Member

Named Gamma-Ray Integrated Detectors (GRID), the mission forms a full-time all-sky one gamma-ray detection network that monitors the transient gamma-ray sky in the multi-messenger astronomy era. Two upgraded CubeSats developed by our NJU team have been successfully launched. I co-led projects analyzing GRBs detected by GRID, including GRB 220408B (Zhang et al., 2023, RAA) and GRB 230812B (Wang et al., 2024, ApJ).

PROFESSIONAL EXPERIENCE/ACTIVITIES

2026 **Oral Talks**, COSPAR 2026, Florence, Italy

- **Oral Talk**, Asia-Pacific Regional IAU Meeting (ARPIM) 2026, Hong Kong, China
- **Oral Talk**, The Third Shenkuo Academic Symposium of the Chinese Astronomical Society, Zhuhai, China

2025 **LOC**, The First HKIAA Symposium, Hong Kong, China

- **Oral Talk**, Einstein Probe: Extragalactic Fast Transients and Multi-messenger Time-domain Astronomy (2025) Workshop, Beijing, China

2024 **Invited Talk**, GRID 2024 Summer Camp, Tsinghua University

- **Teaching Assistant**, Gamma-Ray Bursts: From Devices and Data to Science, Nanjing University
- **Oral Talk**, COSPAR 2024, Busan, Korea
- **Volunteer/Oral Talk/Poster/Ice-breaking Organizer**, The Third Nanjing GRB Conference, Suzhou, China
- **Teaching Assistant**, Gravitational Wave and Related Astrophysics, Nanjing University

REVIEWER/REFEREE

- **Nature Astronomy**
- **The Astrophysical Journal**
- **Journal of High Energy Astrophysics**

SCHOLARSHIPS AND HONORS

2025	Postgraduate Scholarship
2024	National Scholarship for Graduate Student (<i>Most prestigious scholarship for graduate student in China</i>)
2023, 2024	First Class Academic Scholarship
2023	Outstanding Graduate
2021, 2022	The People’s Scholarship of China

PUBLICATIONS

NUMBER OF PAPERS	17
TOTAL CITATIONS	628
H-INDEX	10

An up-to-date paper list on ADS is available on this [link](#).

First Author/Corresponding Author:

- 2026 **Minutes-long soft X-ray prompt emission from a compact object merger.** A. Li, C.-W. Wang, et al., under review, arxiv: 2601.14137
- 2025 **Spectral Hardening Reveals Afterglow Emergence in Long-Duration Fast X-ray Transients: A Case Study of GRB 250404A/EP250404a.** Y.-H. I. Yin, Y. Fang, et al., ApJL, doi: 10.3847/2041-8213/adf552
- 2024 **Triggering the Untriggered: The First Einstein Probe-Detected Gamma-Ray Burst GRB 240219A and Its Implications.** Y.-H. I. Yin, B.-B. Zhang, et al., ApJL, doi: 10.3847/2041-8213/ad8652
- **A Componized Fireball Bubble Fits the Second Extragalactic Magnetar Giant Flare GRB 231115A.** Y.-H. I. Yin, Z. J. Zhang, et al., ApJL, doi: 10.3847/2041-8213/ad2839
- 2023 **GRB 211211A-like Events and How Gravitational Waves May Tell Their Origins.** Y.-H. I. Yin, B.-B. Zhang, et al., ApJL, doi: 10.3847/2041-8213/acfo4a

Selected Co-author:

- 2025 **Evidence for a brief appearance of gamma-ray periodicity after a compact star merger.** R.-C. Chen, B.-B. Zhang, et al., Nature Astronomy, doi: 10.1038/s41550-025-02649-w
- **RapidGBM: An Efficient Tool for Fermi-GBM Visibility Checking and Data Analysis with a Case Study of EP240617a.** Y. Wang, J. Ren, et al., ApJ under review, arXiv: 10.3847/2041-8213/adf552
 - **EP240801a/XRF 240801B: An X-ray Flash Detected by the Einstein Probe and Implications of its Multiband Afterglow** S.-Q. Jiang, D. Xu, et al., ApJL, doi: 10.3847/2041-8213/addebfb
 - **Soft X-ray prompt emission from a high-redshift gamma-ray burst EP240315a** Y. Liu, H. Sun, et al., Nature Astronomy, doi: 10.1038/s41550-024-02449-8
 - **Bridging the Gap: GRB 230812B – A Three-Second Supernova-Associated Burst Detected by the GRID Mission,** C.-Y. Wang, Y.-H. I. Yin, et al., ApJ, doi: 10.3847/1538-4357/ada934
- 2024 **Magnetar emergence in a peculiar gamma-ray burst from a compact star merger.** H. Sun, C.-W. Wang, et al., National Science Review, doi: 10.1093/nsr/nwae401
- **Prospects for detecting neutron star-white dwarf mergers: early warnings from decihertz gravitational-wave observatories.** Y. Kang, C. Liu, et al. MNRAS, doi: 10.1093/mnras/stae340
- 2023 **Synchrotron Radiation Dominates the Extremely Bright GRB 221009A.** J. Yang, X.-H. Zhao, et al., ApJL, doi: 10.3847/2041-8213/acc84b
- **GRB 220408B: A Three-Episode Burst from a Precessing Jet.** Z. Zhang, Y.-H. I. Yin, et al., RAA, doi: 10.1088/1674-4527/acfa59
- 2022 **A long-duration gamma-ray burst with a peculiar origin.** J. Yang, S. Ai, et al, Nature, doi: 10.1038/s41586-022-05403-8
- 2021 **GRB 210121A: A Typical Fireball Burst Detected by Two Small Missions.** X. I. Wang, X. Zheng, et al., ApJ, doi: 10.3847/1538-4357/ac29bd

MEDIA COVERAGE

- **Triggering the Untriggered: The First Einstein Probe-Detected Gamma-Ray Burst 240219A and Its Implications** featured in **AAS Nova Journals Digest**, 31 March 2025.